

CAD File Conversion to GIS Layers: Issues and Solutions

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Abstract

CAD data created for drawings and Geographic Information System (GIS) data are created for different purposes. As a result the software's data models are inherently different, and the files differ in their structure and attributes. CAD data contain information to draw it with features organized by mixed layers. While GIS data has an additional functionality, its feature can hold huge amounts of data describing the features. Moreover, GIS data is "spatially informed" regarding adjacency and other spatial relationships. Thus, the conversion of CAD files to GIS shape file may face problems and data may get corrupted.

Furthermore, in order to keep the geometric properties integrity and accuracy, we should be able to keep a correct topological database. Accurate topological data leads to correct and reliable spatial analysis, in addition to cost reduction. The aim of the topological knowledge in GIS is to increase spatial analysis opportunities. Hence, we should be able to evaluate a topological database, in addition to the geometric properties.

Different issues will be addressed in this work; a step-by-step methodology will be presented using a case study to demonstrate the problems, and provided solutions using different software and programs. Examples of problems and solutions when transferring CAD data into GIS data layers are provided to allow correct and full functionality of the GIS.

Key words: Environmental Systems Research Institute (ESRI), CAD, GIS, Topology, Shape Files, Geodatabase.

Introduction

The increasing demand for the data and information of a spatial nature in the past three decades, has led to the development of automated tools for efficient storage, analysis and presentation of geographic data. This rapidly evolving technology has come to be known as "Geographic Information Systems (GIS)". Geographically referenced data separates GIS from other information systems. In order to describe a road using GIS, we refer to its location (spatial data) and characteristics (data attributes: length, name, speed limit, etc.). GIS import data from many types of sources. This presents a great challenge because these sources have data structures that differ from GIS data structure. The most important source of data for GIS is the Computer-Aided Design (CAD) or Computer Aided Drafting (CAD) files. Thus, importing data from these sources requires some necessary treatment and management such as building topology, modifying the attributes, editing the spatial features, and separating and classifying of layers. Recent advances in GIS software have made it simpler to move data

into GIS shape files or geodatabases, however, many quirks still exist in file transfer due to the intrinsic differences in the file formats and structure. There are not many works that deal with the data transfer compatibility, integrity and accuracy to allow better GIS functioning.

The NPS developed a step-by-step method for transfer; noted issues in both the CAD and GIS to be aware of and created GIS-friendly guidelines for outsourcing new CAD work. The produced document is intended to be a guide, but it doesn't tackle every technical issue and likely situation in CAD to GIS transfer.

Data migration is often the most challenging and expensive step in the GIS implementation. The key here is the data accuracy and specification generated for the data model before commencing the full construction. Data validation in both CAD and GIS is crucial to the realization of data movement and validation routines need to be devised. These routines are normally developed in CAD using Visual Basic and in GIS using Acute Myeloid Leukemia (AML). Xiaodong [1] built a

Geodatabase into a Multi-user GIS; migrating CAD Data to a Geodatabase, and GIS. Linyan et al. [2] clarified how to overcome some types of problems in the conversion of CAD data into GIS for land planning purposes. Both dealt with the integrity on statistical and numbers level. They delineated a methodology which creates the desired end result: GIS integration with Building Automation Systems, Computerized Maintenance Management Systems, Energy Management Systems and Enterprise Asset Management. The BIM and CAD data with robust GIS systems and techniques have produced a new paradigm that gives facility to interested users of GIS; an access to normalized data resources. The problem of editing was resolved in most GIS software systems. The Geodatabase, through ArcSDE in ArcGIS supports multi-user access, editing, versioning and rollback relying on a relational database structure. The Geodatabase allows users to define objects and behaviors [3]. A report produced by ESRI – Infograph [4], emphasized the importance of topology for the editing process, but it didn't address every technical issue and possible scenario in CAD to GIS transfers. In 2008, we interpreted the problem of data importing or converting and its effect on the quality and the integrity of the data [5].

CAD and GIS data are created for different purposes. Thus the software's data models are inherently different, and the files differ in their construction and attributes. A CAD file deals with points, lines, or polylines and does not contain additional attributes. It contains only the information needed to draw it (line weight, type, etc.). GIS data has an additional functionality; its feature can hold huge amounts of data describing the features. GIS data is "spatially informed" regarding adjacency and other spatial relationships. Other main difference in CAD data is that features are organized by mixed layers, which can contain points, lines, polygons, and annotation as compared with a GIS shape file, which can only contain one feature type. When a CAD drawing is exported to a GIS shape file, the CAD drawing's layers become a field in the GIS shape files attribute table. New point, line, and polygon shapefiles can then be obtained based on each layer (trees, buildings, roads, etc.). The main objective of this work is to define a methodology for resolving the unsolved problems of conversion from CAD files to GIS shapefiles while keeping data integrity and GIS functionality. For this purpose a real application of conversion CAD to GIS shapefiles will be presented and the site

of Al-Balqa Applied University (BAU) is chosen to be the site of application.

The Methodology and Application

Different Software and programs should be used in the data processing (each handles the converted data differently) such as: (a) Arc GIS, (b) Geomedia Professional, (c) AutoCad, (d) ASP.Net and (e) ETGeoWizards,

Data Collection: A GIS and a database system are built for the BAU using field collected data about the employees, the location of their offices in the buildings and the plans for the buildings floor by floor in CAD files; see Figures 1 and 2.

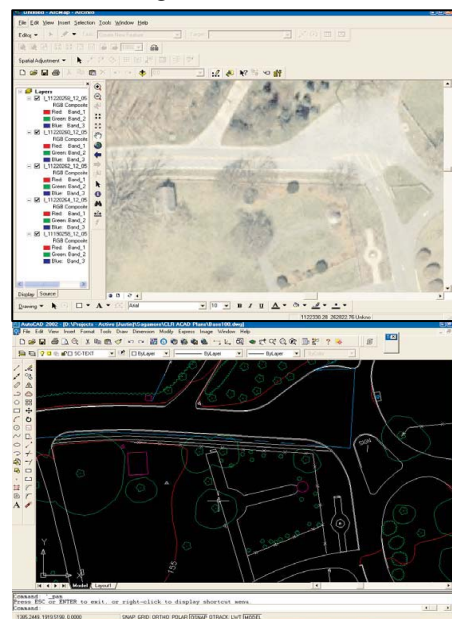


Figure1: A Sample of BAU (CAD files) Raster

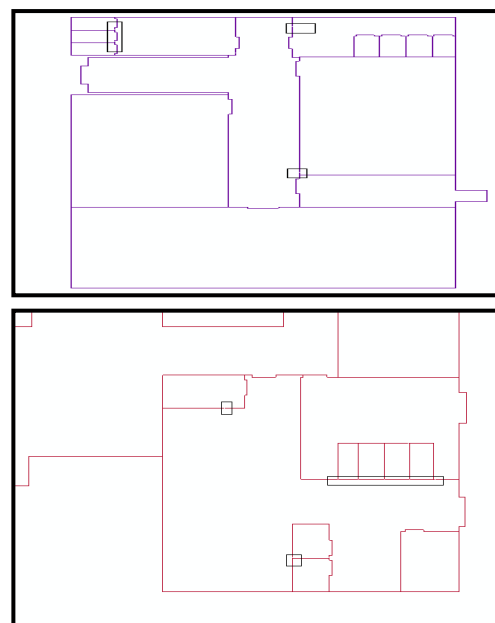


Figure 2: Floor Plans for BAU (CAD files) vector

Creating GIS Shapefiles from CAD layers

When a CAD drawing is imported into GIS, it will interpret all elements into the four primitive geometric GIS features: points, lines, polygons, in addition to annotation into shapefiles or feature classes. This division is based on the original CAD layers with no interaction between data and no connection to the real world or to other design objects. GIS contain two types of data: (a) raster data (two-dimensional arrays of data values), such as those generated by satellites and other remote sensing operations; and (b) vector data (lines and polygons). One of the most popular CAD packages used in industry sectors is Autodesk's AutoCAD. However, joining AutoCAD designs and corporate mapping databases is currently not an easy task. Hence, an AutoCAD user has to find some way of translating mapping data into DXF (AutoCAD format) files, and then store these files somewhere for use. During the translation process, the source data may have been updated in a way that can affect the accurate depiction of the design [2].

In order to converge CAD files to GIS shapefiles we used different GIS software tools to allow better debugging and error detecting such as: Geomedia software, ArcGIS software and, CAD 2 shape program

The Use of Geomedia Professional Software:

This product allows combining geographic data from different sources, in different formats, and with different map projections; all into a single environment. It permits performing composite queries on spatial data from various sources, and produces numerous views of highly sophisticated maps in a single GeoWorkspace.

The applicability of a GIS project directly depends on the accuracy and topological consistency of the vector data that form the basis for GIS. The existence of errors or inconsistency in the vector data could easily avoid a reliable GIS analysis. Therefore, it is essential to do an overall check as well as to compare them in terms of integrity in order to eliminate the errors coming from the encoding process.

All the AutoCAD drawings for all the features in the study area (building, and streets) were transferred to GIS feature classes or shapefiles as described below.

- All closed polylines should be converted to polygons.
- Need to create attributes for the transferred polylines
- For advanced polygon creation, we have to use the Build Polygon Wizard.
- Separating the layers depending upon their geometric feature.
- Apply the editing process, which will involve intercepting errors and modifying features; the attribute data will be built and transferred to the corresponding polygon features (linked to their spatial features).

Using Topological Data Structure to Resolve Problems of Conversion

Topology examines the characteristics such as neighborhood and merges outside the coordinate system information of geometric objects [3, 7, 8]. It is the geometric relationship between edges, nodes and the faces and a way in which logical relations can be defined (neighborhood, coincidence, inclusion, intersection, and sharing), in addition to metric relations (coordinate, length and area) [9]. The aim of the topological knowledge in GIS is to increase spatial analysis opportunities.

In order to represent the spatial information as well as spatial relations (neighborhood, coincidence, directions, and links) of the features in topological data structures on the computer; the node elements corresponding to point, and edge (arc) elements corresponding to the lines and the face elements corresponding to the polygons are used. Note that more than two line intersection points may be an exposed end of the line or a single point not connected to any side. Edge is a set of coordinate pairs starting with a node and ending with a node. Face is the largest two-dimensional space restricted by the edges and cannot be divided by an edge.

According to this, point features are composed of nodes, line features with one or more edges, and the face features surrounded by one or more edges. To be able to evaluate a topological database, in addition to the geometric properties, the following relationships must be determined and stored:

- (1) Edges making up the boundaries of each polygon

- (2) Neighborhood relations between the polygons
- (3) Connections at the intersection points
- (4) Start and end points of edges

Accurate topological data leads to correct and reliable spatial analysis, in addition to cost reduction. In the databases where topological properties can be stored and used, information about a line feature is connected to another line feature, depending on where, and at what point line features are combined. The polygon features to the right and left line features can be easily and quickly queried. The existence of the topological data in addition to the spatial data and attribute data in database will cause the increase in volume of the database. The advantage of topological data that allows spatial analysis and queries expected from GIS offers very significant advantage compared to data storage disadvantages.

Topological errors in spatial vector data and its visualization and elimination by means of AutoCAD Vector auto lisp as follows. Spatial vector data contain many topological errors arising from: (a) the quality of the source material, (b) data collection techniques and (c) the human interaction. Acquisition of vector data for GIS requires further processing in order to be used for advanced spatial analysis. Before using spatial data for analysis, it must be made topologically-correct. Vector features can be made to respect spatial integrity through the application of topology rules such as polygons must not overlap. To make data usable, some quality checks should be applied. All GIS software packages have the functionality to perform this kind of task and also capable of providing the spatial integrity and cleanness during data capturing. The topological errors arising from the conversion of data into the digital maps could be easily removed through the software written in AutoCAD LISP.

Results and Discussion

The following method was followed for processing the data conversion in our application at BAU.

Adding CAD files to a Dataview Exporting CAD data as Shapefiles:

CAD drawings are immediately displayed in ArcGIS (dataview) but, they needed to be converted to GIS layers. When a CAD file is added to GIS software, it is divided into data sets, based on the four types of GIS data: point, line, polygon, and annotation. It is to be noted that exported CAD files will be in two

types of files. The annotation set cannot be exported as a shapefile. Annotation must be exported as a feature class of a geodatabase. The exported files from CAD to GIS layers will be separated in to layers having the same primitive geometric features: points, polylines or polygons.

In order for newly created shapefiles to be compatible properly with other GIS data, they need to meet the following criteria:

(a) they have a coordinate system projection defined in GIS. The Universal Transverse Mercator (UTM) coordinate system was chosen the standard datum and it will be NAD83 for all layers in this application. To determine if a shape file has a defined coordinate system, we have to check the original CAD drawing notes, compare coordinates of an identical spot in CAD and GIS, and/or check with the CAD drawing developer.

Another method to determine the coordinate system consists of adding the file to a blank document without any information. Then, we give a system of projection to this layer. Some layers are given a system of projection, but the overlay with other layers doesn't properly occur (overlaid). In this case, it requires 4 correctly distributed points (at least) to be defined in x and y coordinates. Then we have to perform the geometric correction. This process will move the shapefile from its current coordinates to the correct coordinates.

Another problem we faced was that the original CAD drawings were created using a defined coordinate system not recognized by the GIS software. In this case, we specified the parameters in the "Define Projection Wizard" and then we had to spatially adjust the shapefile by using the similarity or the affine transformation, which depends on the type of errors. The Similarity Transformation maintains the aspect ratio because it does not differentially scale or skew the obtained data. If this algorithm does not perform well, we have to choose "Affine transformation". Nevertheless, the affine transformation will differentially scale, skew and rotate translated data. The remaining adjustment methods were often too aggressive to be useful for CAD data adjustments. In this way, the spatial adjustment can be executed and we have to complete the adjustment. When a satisfactory match has been achieved, we have to stop the editing process. The shapefiles are now properly overlaid with other UTM NAD83 data. The obtained residuals were satisfied (0, 0012 meter).

(b) to repair specific topological errors in the vector data of our application such as: (1) Floating or short lines, (2) Overlapping lines, (3) Overshoots and undershoots and (4) Unclosed and weird polygons.

Before transferring the CAD drawing to GIS form, we have to verify that all features are on their respective layers. We have to isolate each layer by turning all other layers OFF and to see that there are no stray drawing elements or features that belong on other layers.

Removing Duplicating Arcs: According to topological integrity rules, lines must not overlap and duplicating lines must be removed to avoid obtaining false results when we do spatial analysis; see Figure 3.

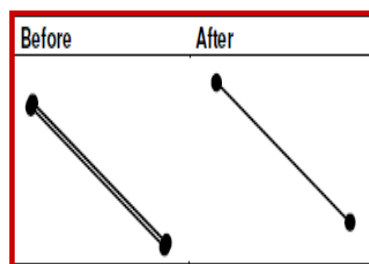


Figure 3: An example of removing a duplicated layer.

Correcting the Errors of Overshoot and Under Shoot: The errors of overshoot and undershoot create a great problem for GIS analysis and give fault results for networking processing (the real shortest distance). To correct this type of errors, we use the editing process in GIS; see Figure 4.

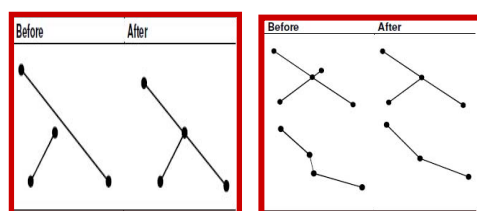


Figure 4: Correcting undershoot and overshoot errors.

Correcting Opened Polygons: This type of errors forms a great challenge for the correct conversion of layers from CAD to GIS layers. Many errors in the data base were marked; see Figure 5.

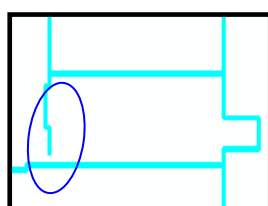


Figure 5: An unclosed polygon detected

(c) To Create new shapefiles from CAD files using the attribute table: When a CAD drawing is converted to a GIS shapefile, the original CAD layers become a field in the shapefile's attribute table. By selecting objects in the shapefile based on their CAD layer names, and exporting them as new shapefiles, shapefiles can be created for each original CAD layer. Therefore, this procedure can yield separate shapefiles for grouping objects (trees, buildings, roads, etc.) which can then be integrated more effectively into a GIS database system.

Correcting topological errors in our application: This step is very important for obtaining a reliable correct database by modifying and correcting all the topological errors. The most important types of topological errors consist of neighborhood, adjacency and relationships between different database. Two polygons shared an arc must be corrected from the sliver or splinters errors; see Figure 6. We note that splinters or slivers are thin polygons between two adjacent polygons due to encoding errors and must be eliminated in order to obtain a reliable, complete and correct database.

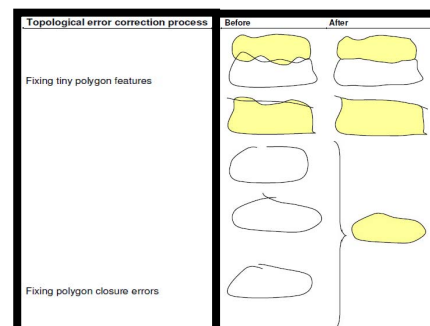


Figure6: Topological errors

To add attribute data: CAD files are not very efficient to link non spatial data to their spatial data. Hence, for all features we have to build attribute data. In addition, GIS data must be organized and managed in different ways in the data base. For example: there are no associate tables in CAD files and there is no capability to do data fusion or other operations of data management as in GIS software.

Conclusion

The process of integrating different types of data into GIS is very important these days as different types of software perform different functions. More and more functions and capabilities that need to be more friendly to users require the conversion of the data while

keeping its integrity intact. CAD Autodesk offers integrated CAD and GIS solutions for surveying, mapping, civil engineering, city planning, and infrastructure lifecycle management, however, there are still some problems we encountered while transferring the data.

This work reported here outlines a methodology for the transfer from CAD files (butter drawing and 3D modeling capabilities) to GIS database. It requires necessary treatment and management such as building topology, modifying the attributes, editing the spatial features, separating and classifying of layers. It requires processing and editing to build complete, efficient and correct database. This is important in order to obtain the full functionality to perform the different operations of overlay, network analysis or any other GIS operations. These operations are necessary due to the difference in the structure of various related software tools and packages. These solutions create, integrate, and manage CAD and GIS features with no loss of precision, topology or functionality.

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